Lean formalization of an Assumptions of Physics technical brief: log

This log records which statements of the technical brief in this directory (`https://assumptionsofphysics.org/resources/briefs`) were checked in Lean, where the formal version differs from the brief, and any problems found in the brief.

Setup

- Lean `v4.36.0-rc1`; Mathlib at commit `2cabef97`; physlib (`master`) and TauCeti (commit `eb4dd24`, on the same Mathlib) as further dependencies.
- Build: `lake build` in the project root.
- Every file is checked with `#print axioms` on its main results. A result is recorded here only if it depends on no axioms beyond `propext`, `Classical.choice` and `Quot.sound`, and so has no `sorry`.
- Imports are kept to the Mathlib modules actually needed.
- Material that belongs in Mathlib is kept in `ForMathlib/`, in Mathlib's directory layout (library `ForMathlib`), and imported by the brief files.

1 Non-uniqueness of the moment problem for Schwartz functions

Files.

- File: `briefs/SchwartzMomentProblem/SchwartzMomentProblem.lean`
- Library: `SchwartzMomentProblem`
- Imports: `Mathlib.Analysis.Calculus.BumpFunction.InnerProduct`, `Mathlib.Analysis.Distribution.SchwartzSpace.Basic`, `Mathlib.MeasureTheory.Measure.Lebesgue.Basic`

Setting. Schwartz functions $f, g \in \mathcal{S}(\mathbb{R}, \mathbb{C})$ (`SchwartzMap`) with compact supports and $\operatorname{tsupp} f \cap \operatorname{tsupp} g = \emptyset$, and $\psi_\theta = f + e^{i\theta} g$. The moments are

$$\langle \psi | X^m P^n | \psi \rangle = \int_{\mathbb{R}} \overline{\psi(x)}\, x^m\, (-i\hbar)^n\, \psi^{(n)}(x)\, dx,$$

with $\hbar$ an arbitrary real parameter.

Results.

Brief	Lean
cross terms vanish, pointwise eq. (3): $\langle\psi_\theta\|X^m P^n\|\psi_\theta\rangle = \langle f\|X^m P^n\|f\rangle + \langle g\|X^m P^n\|g\rangle$	`conj_mul_iteratedDeriv_eq_zero`, `density_psi` `moment_psi`
each θ gives a different state	`linearIndependent_psi`: $\psi_\theta, \psi_{\theta'}$ linearly independent when $e^{i\theta} \neq e^{i\theta'}$
the bump functions of Fig. 1	`bump`, `tsupport_bump_subset`, `bump_ne_zero`
Theorem 1	`exists_same_moments`, with $\psi_1 = \psi_0$, $\psi_2 = \psi_\pi$ for bumps centred at 0 and 3

Differences from the brief.

- The moment is defined as an integral for every Schwartz function, but the identity for ψ_θ is proved only under the brief's hypothesis that f and g have compact support. This makes every integrand continuous with compact support, hence integrable.
- "Each θ defines a different physical state" is formalized as linear independence of ψ_θ and $\psi_{\theta'}$, which holds exactly when $e^{i\theta} \neq e^{i\theta'}$, i.e. for $\theta \not\equiv \theta' \pmod{2\pi}$.
- The closing remark on Schwartz seminorms and the conclusion about local operators are not formalized.

Problems found in the brief. No mathematical problems. The argument needs f and g to be nonzero, which the brief leaves implicit, and "different state for each θ" holds only modulo 2π.